

Simulating Human Opinions with Large Language Models

Opportunities and Challenges for Personalized Survey Data Modeling

Carolin Kaiser
Nuremberg Institute for Market
Decisions
Nuremberg, Germany
carolin.kaiser@nim.org

Jakob Kaiser
Nuremberg Institute for Market
Decisions
Nuremberg, Germany
jakob.kaiser@nim.org

Vladimir Manewitsch
Nuremberg Institute for Market
Decisions
Nuremberg, Germany
Vladimir.Manewitsch@nim.org

Lea Rau
Ludwig-Maximilians-University
Munich, Germany
rau@lmu.de

Rene Schallner
Nuremberg Institute for Market
Decisions
Nuremberg, Germany
rene@renerocks.ai

Abstract

Public and private organizations rely on opinion surveys to inform business and policy decisions. Yet, empirical surveys are costly and time-consuming. Recent advances in large language models (LLMs) have sparked interest in generating synthetic survey data, i.e., simulated answers based on target demographics, as an alternative to real human data. But how well can LLMs replicate human opinions? In this ongoing project, we develop and critically evaluate methods for synthetic survey sampling. As an empirical benchmark, we collected responses from a representative U.S. sample ($n = 461$) on preferences for a common consumer good (soft drinks). Then, we developed ASPIRE (Automated Synthetic Persona Interview and Response Engine), a tool that pairs each human participant with a “digital twin” based on their demographic profile and generates synthetic responses via LLM technology. Synthetic data achieved better-than-chance accuracy in matching human responses and approximated aggregate subjective rankings for both binary and Likert-scale items. However, LLM-simulated data overestimated humans’ tendencies to provide positive ratings and exhibited substantially reduced variance compared to real data. The match of synthetic and real data was not systematically related to participants’ age, gender, or ethnicity, indicating no demographic bias. Overall, while synthetic sampling shows promise for modeling aggregate opinion trends, it currently falls short in replicating the variability and complexity of real human opinions. We discuss insights of our ongoing project for accurate and responsible user opinion modeling via LLMs.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Human computer interaction (HCI);

HCI design and evaluation methods; User models; • **Applied computing** → Operations research; Marketing.

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1 Introduction

Businesses and public organizations rely on accurate insights into citizens’ attitudes and interests to make decisions that align with public needs. Traditionally, such insights are obtained through empirical surveys. However, large-scale surveys are often costly and time-consuming. Moreover, declining response rates in human surveys raise concerns about data quality and non-response bias [13, 20]. These challenges highlight the need for scalable, cost-efficient alternatives to model public opinions, especially when faced with limited resources or time-sensitive decisions.

Recent advances in large language models (LLMs) offer promising opportunities for human opinion modeling. It has been proposed that LLMs can simulate human survey responses by conditioning the model on a persona (e.g., demographic profile) and prompting it to answer as if it were that person [15, 25]. The resulting outputs form synthetic datasets generated by AI-driven “virtual participants.” This approach has been described using various terms, including “synthetic user,” [16] “synthetic consumer” [9], “homo silicus” [18], and “silicon participant” [3]. In this paper, we refer to it as synthetic sampling. If reliable, synthetic sampling could offer a compelling supplement to traditional surveys by enabling fast, low-cost simulations of human responses [2, 7, 11]. As a result, the approach has attracted interest across different scientific disciplines [1, 2, 17, 21, 26].

However, findings from previous work are mixed. While some studies report that LLM-generated responses approximate aggregate human opinions, others highlight limitations, including a failure to capture nuanced differences in views between humans or the

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presence of demographic biases [4, 8, 11, 19]. These concerns underscore the need for a rigorous evaluation before synthetic sampling can supplement conventional survey methods.

Fueled by the promise of fast and cost-effective public opinion models, interest in synthetic sampling has risen in the domain of professional consumer market research [5]. Market research is crucial for companies to understand consumer behavior, e.g., by tracking brand perceptions and identifying or predicting consumer needs. Yet, since most prior validation efforts of synthetic sampling have focused on political or psychological topics, its applicability for practical consumer research is unclear. Moreover, public surveys can use different question formats to gauge public opinions, such as categorical (choice-based) or numerical (rating-based) questions. Currently, however, it is unclear whether LLM accuracy depends on question format. Determining this would be important to inform the design of studies that intend to employ synthetic sampling.

Our current project examines the potential and limitations of LLMs to simulate consumer survey responses. We focus on a practical scenario: Can general-purpose LLMs predict individual preferences for consumer products based on their demographic data? To investigate this, we collected real survey responses from a representative U.S. sample on their preferences for an everyday consumer good (soft drinks) and generated corresponding synthetic responses using ASPIRE, a custom-built tool for synthetic sampling. For each real participant, we generated a “digital twin”, a persona mirroring their demographic features, and the LLM was prompted to complete the survey from that perspective. We statistically evaluated whether the synthetically produced data could replicate both overall trends and individual-level variability. By including both binary and numerical questions, we also tested whether the accuracy of synthetic responses depends on survey format. As a work-in-progress, our study offers insights into the potential and limitations of systems that model user and consumer preferences via generative AI.

2 Methods

2.1 Empirical Sample

As an empirical benchmark for our synthetic survey, we conducted an online survey with 492 participants from the United States. The sample was quota-matched to the U.S. population in terms of age, gender, and ethnicity. After excluding individuals who either did not complete the survey or reported never purchasing soft drinks (the focal topic in our study), the final sample comprised 461 participants (mean age = 46.4 years, range = 18–82, 52.8 % female, 47.0 % male) from 48 different U.S. states. The study was conducted via Prolific and QuestionPro.

2.2 Survey Design

Our goal was to evaluate the suitability of LLM-based tools for simulating human attitudes on topics commonly addressed in professional market research. To this end, we designed a questionnaire assessing participants’ attitudes toward various soft drink brands as a product category with which most American consumers are likely familiar. We selected four widely recognized brands (Coca-Cola, Pepsi, Sprite, 7UP) and four significantly less well-known brands in the U.S. (Dry, Moxie, Blue Sky, Orangina), based on a recent market survey [27]. This allowed us to test whether simulated data could

differentiate between responses to popular and less familiar brands. The survey was based on questionnaires commonly employed in practical consumer market research, based on the frequently used marketing funnel model [29]. Participants answered questions on brand awareness (e.g., “Which of these brands are you aware of?”), brand consideration (e.g., “Which of these brands do you generally consider when buying soft drinks?”), and brand purchase (e.g., “Which of the following soft drink brands do you currently purchase?”). For each question, participants had the option to either select or not select individual brands, resulting in 24 binary choices.

In addition, for each brand that participants indicated as familiar, they evaluated that brand via Likert-scale ratings (1–7) in four questions per brand, covering several aspects like perceived quality or their likelihood to recommend the brand to friends or colleagues. Thus, this design enabled us to collect both binary and continuous measures of consumer attitudes, reflecting common question formats used in market research for assessing opinions about both well-known and niche brands of everyday products.

2.3 Synthetic Sample

To automate the generation of synthetic survey data, we developed ASPIRE (Automated Synthetic Persona Interview and Response Engine), a new Python-based tool that combines two core components: (1) a questionnaire interpreter capable of parsing a textual survey, and (2) a structured interview module that simulates responses using LLMs conditioned on synthetic personas. ASPIRE was implemented in Python 3.11 and interfaces with OpenAI’s GPT-4o model via the official OpenAI Python library. API calls were made to the ChatCompletion endpoint using default sampling parameters (temperature = 1.0, top_p = 1.0). Note that, while the current implementation employs the OpenAI API, ASPIRE can in principle be adapted to other LLM APIs.

Synthetic personas were created by mirroring the sociodemographic attributes of real survey participants. Each respondent was assigned a synthetic counterpart with the same characteristics, derived from our empirical data. The following attributes were included in the persona definition: gender, year of birth, education level, occupation, financial situation (as a self-assessment on a 5-point Likert-scale), U.S. state of residence, ethnicity, country of birth, nationality, and native language. To establish context, GPT received a system prompt that included the original survey welcome text, followed by a structured persona description in the following format:

You are a [ethnicity] [sex] living in [state]. You were born in the year [year] in [country of birth]. Your nationality is [nationality] and you speak [language] fluently. Your education level is: [education level] and your profession is: [profession]. You describe your financial situation with [financial situation].

The questionnaire interpreter then sequentially prompted the model to answer each survey question, dynamically adapting the flow based on prior responses. Conditional logic was automatically handled by interpreting rules encoded in the survey structure. For example, the current survey necessitated that participants were only asked to numerically rate brands that they previously identified as familiar. This filtering was performed automatically by ASPIRE. To

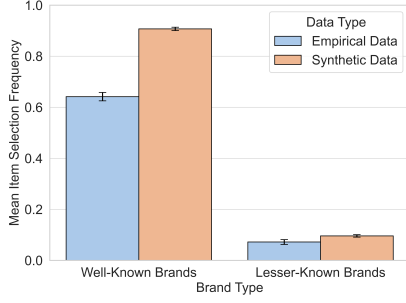


Figure 1: Average item selection frequency on binary survey items measuring brand familiarity in real and synthetic data.

ensure structured and machine-readable output, we used the API’s JSON Mode and explicitly specified the expected schema for each question.

3 Results

3.1 Binary Survey Items Simulation

We evaluated real participants’ tendency to select brands in the survey section that asked for familiarity of different brands and compared it with the selection tendency of the synthetic participants (cf. Figure 1) via an ANOVA with the factors BRAND TYPE (well-known/lesser-known) and DATA TYPE (empirical/synthetic). This revealed a main effect of BRAND TYPE, $F(1,446) = 16475.49$, $p < .001$, $\eta^2_p = 0.91$, a main effect of DATA TYPE, $F(1,446) = 793.33$, $p < .001$, $\eta^2_p = 0.30$, and a BRAND TYPE $\times$ DATA TYPE interaction, $F(1,446) = 576.20$, $p < .001$, $\eta^2_p = 0.23$. As expected, participants were more likely to express familiarity for well-known brands than for lesser-known brands, $p < .001$, $d = 3.92$. The simulated data correctly captured the differentiation between popular and unpopular brands, $p < .001$, $d = 12.96$. However, as indicated by the main effect of DATA TYPE, the synthetic survey significantly deviated from the empirical data. The synthetic data overestimated the selection frequency for survey items both for popular brands, $p < .001$, $d = 1.96$, and unpopular brands, $p < .001$, $d = 0.29$. As indicated by the BRAND TYPE $\times$ DATA TYPE interaction, the tendency to overestimate selection frequency was significantly higher for well-known compared to lesser-known brands. Thus, while synthetic data correctly differentiated between more or less well-known brands similar to the real participants, synthetic responses were biased towards overestimating participants’ tendency to select items (i.e., brands) in the survey. On average, the agreement between real and corresponding synthetic participants regarding their binary selection was 78.64 % (SD = 8.68), which was significantly better than chance, $p < .001$, $d = 3.30$.

3.2 Numeric Rating Items Simulation

We compared the numerical ratings by human participants with simulated rating scores of the synthetic sample (Figure 2). Because participants only rated brands they had previously indicated familiarity with, this analysis includes only those who recognized at least one brand from each category ($n = 201$). We found a main

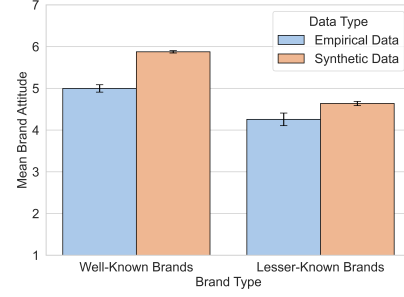


Figure 2: Average rating tendency on numerical item ratings for empirical and synthetic data.

effect of BRAND TYPE, $F(1,200) = 453.98$, $p < .001$, $\eta^2_p = 0.22$, a main effect of DATA TYPE, $F(1,200) = 97.40$, $p < .001$, $\eta^2_p = 0.15$, and a BRAND TYPE $\times$ DATA TYPE interaction, $F(1,200) = 32.99$, $p < .001$, $\eta^2_p = 0.02$. As expected, well-known brands received more positive ratings than lesser-known brands by real participants, $p < .001$, $d = 0.61$. This rating tendency was correctly replicated in the synthetic sample, $p < .001$, $d = 1.20$. However, as the DATA TYPE main effect revealed, synthetic ratings showed significant deviations from the real ratings. Synthetic data overestimated humans’ positive attitudes for both well-known brands, $p < .001$, $d = 1.19$, and lesser-known brands, $p < .001$, $d = 0.52$. As indicated by the BRAND TYPE $\times$ DATA TYPE interaction, this overestimation was higher for well-known than lesser-known brands. The average agreement rate between real and synthetic data was 78.65 % (SD = 11.20), which was significantly higher than expected by chance, $p < .001$, $d = 0.65$.

3.3 Real vs. Synthetic Data Variability

While one important aspect of synthetic data is to approximate the central tendencies of real populations, it would also be important that synthetic data reflects the diversity in opinions between different people. Figure 3 shows data distributions for averaged binary and numerical response tendencies. We employed Brown-Forsythe-Tests to compare the variance between real and synthetic data. We found that synthetic compared to real data had significantly lower variability for both average binary response behavior, $p < .001$, $\eta^2 = 0.185$, as well as for numerical rating tendencies, $p < .001$, $\eta^2 = 0.029$. In both cases, the mismatch in variability was stronger for well-known brands than for lesser-known brands (all p ’s $< .001$, all $\eta^2 > .08$). Thus, the LLM-generated data simulation led to a more uniform response pattern, in contrast to more variability between human participants.

3.4 Sociodemographic Bias in Data Simulations

One concern in the generation of synthetic data with LLMs is the potential for sociodemographic bias, meaning systematically lower accuracy for certain population subgroups, e.g., due to limitations or imbalances in the LLM’s training data [6, 24]. To test this, we conducted multiple regression analysis predicting the match accuracy between real and synthetic responses of each participant with their sociodemographic characteristics (year of birth, gender,

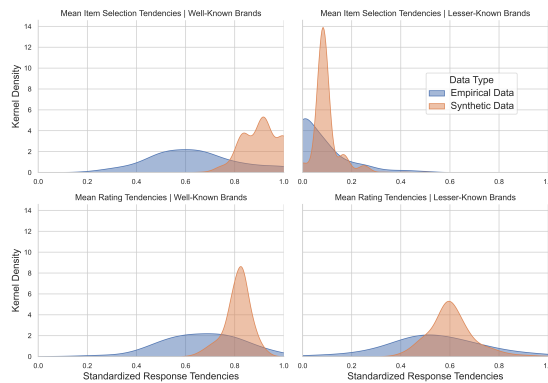


Figure 3: Density distributions of mean response tendencies for empirical vs. synthetic data, standardized to 0-1 to aid comparison. Top: binary selections, with 1 = 100% of items selected. Below: Likert ratings, with 1 = maximum rating given.

ethnicity, education, profession, and self-reported financial situation) as predictors. None of these variables significantly predicted accuracy (all p 's > .10), showing no evidence of sociodemographic bias in our U.S. sample.

4 Discussion

This study explored the capabilities and limitations of LLMs in simulating human survey responses as commonly needed in professional market research, using personalized “digital twins” to generate synthetic samples. The results highlight both the potential and current considerable constraints of synthetic sampling via LLMs.

Our approach achieved above-chance accuracy in predicting individual responses and reproduced aggregate trends in consumer attitudes. Specifically, the synthetic data reflected broad differences in brand familiarity and preference between well-known and lesser-known consumer goods. These findings suggest that, when provided with basic demographic profiles, LLMs can generate synthetic responses that broadly approximate real consumer opinions. However, the results also revealed that synthetic responses systematically overestimated positive consumer attitudes, particularly toward well-known brands. Additionally, the variance in responses among synthetic participants with different persona profiles was significantly lower than in the responses provided by real participants. This indicates a lack of individual differentiation. This pattern was observed across both binary response selections and Likert-scale ratings, suggesting that the limitations are not dependent on the question format. This indicates that LLMs tend to overgeneralize or default to socially desirable responses, which limits their ability to capture the diversity and nuance of human opinions. As a result, synthetic survey data may currently be most appropriate for early-stage concept testing or low-stakes applications, rather than for use cases that require precision or detailed insights.

We did not find biases in simulation accuracy based on sociodemographic factors. Crucially, the absence of such effects in our

data does not imply that LLM-generated data is generally bias-free. The topic of our survey (consumer preferences for soft drinks) was relatively benign, which likely reduced the risk of demographic stereotyping. In more sensitive domains such as health, politics, or medicine, the risks of bias may be considerably higher. Moreover, prior research indicates that answers generated by LLMs like ChatGPT align more with American cultural norms but might exhibit limited adaptability to other cultural contexts [10, 28]. Thus, applying synthetic sampling methods in high-stakes contexts necessitates a high degree of caution.

Looking ahead, several strategies may help improve the representativeness of LLM-generated opinion data. One approach might be to incorporate more detailed information about the individuals being modeled, such as psychological traits or domain-relevant behavioral patterns. An important open question is which types of individual-level data will be most valuable for enhancing synthetic accuracy [22, 23]. In addition, integrating techniques such as retrieval-augmented generation (RAG) could support the production of more context-sensitive outputs. RAG enhances LLMs by integrating external contextual data, such as economic indicators or product reviews in the domain of consumer research, into the generation process [12, 14]. By enriching LLM prompts with relevant background information, RAG may help tailor synthetic responses more precisely to specific survey contexts. We are currently working to extend ASPIRE with such capabilities and to evaluate their impact empirically. In conclusion, while our findings highlight the potential of LLM-generated synthetic data, they also underscore its current limitations in capturing the complexity and diversity of human opinions. As a work-in-progress, our project contributes to the development of AI-based tools that support personalized user modeling and can inform the design of adaptive systems relying on accurate approximations of human evaluations.

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